

Power-Kernel Multiplicative VCG for Multi-Agent Failure Attribution:

A Unified Family with Trace-Length-Adaptive Selection

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Executive Summary

Main result. Within the multiplicative-VCG framework of the companion theory note, we propose a one-parameter kernel family

$$\kappa_{c,p}(x) := 1 - \min(|x|^p, c^p), \quad p \geq 1,$$

which subsumes both the original squared kernel ($p = 2$) and the L1-Huber kernel ($p = 1$) as special cases. A single trace-length-based selector picks the kernel parameters from a two-element menu, giving a single deployable mechanism that improves over the original VCG on both Who&When subsets. Table 1 reports the headline numbers.

Mechanism positioning. The mechanism is a *static, single-shot* VCG mechanism: it consists of the allocation rule (Gauss–Seidel solving the social-welfare objective) and the Clarke pivot payment rule. We report results with $R = 0$ OMD rounds, that is, no iterative refinement of the discriminator reports. This is deliberate: we provide a **two-layer equilibrium guarantee**,

- *report-level equilibrium* (Theorem 2): under the Clarke pivot payment, truthful reporting is a dominant-strategy equilibrium for every kernel $\kappa_{c,p}$;
- *allocation-level equilibrium* (Proposition 3): the Gauss–Seidel output $\mathbf{d}^*$ is a coordinate-wise Nash equilibrium of the social-welfare game, satisfying the joint M-estimator FOC at every step.

Both equilibria are *static* properties of the mechanism. They do not require an outer OMD calibration loop. We empirically investigate OMD calibration with three objectives (payment, allocation entropy, top-1 margin) and report in Section 4 that all three degrade step accuracy on AG due to a consensus-drift mechanism. The single-shot two-layer equilibrium delivers the inference task on its own.

Table 1: Step-attribution accuracy on Who&When. **Bold** marks the proposed method; the original VCG row uses Tong’s reported configuration $R = 5$ OMD rounds, $\varepsilon = 0.10$ external clip (Section 3.1). Best-single is the oracle ceiling (requires ground-truth selection).

Subset	n	Single discriminator (<i>oracle reference</i>)			Aggregation / mechanism (<i>practical</i>)			
		Gemini	Qwen	GPT-OSS	Mean	Orig. VCG	L1-Huber	Power-VCG
HC	55	14.5%	10.9%	16.4%	12.7%	9.1%	23.6%	23.6%
AG	126	38.1%	27.0%	19.0%	33.3%	34.9%	30.2%	36.5%

Why a kernel family rather than a single kernel. Two empirical observations motivated the family:

1. The L1-Huber kernel ($p = 1, c = 0.05, \varepsilon = 10^{-6}$, no OMD) significantly improves on HC: +14.5pp over the original VCG, McNemar $p = 0.022$. The cross-step reputation-weighted median update recovers signal that per-step aggregators miss; the gain is concentrated in the maximal-disagreement regime (60% of HC traces).
2. On AG, however, L1-Huber does not match the original squared VCG. A (p, c) scan at the appropriate clipping level ($\varepsilon = 0.10$) reveals a non-trivial optimum at $(p = 1.75, c = 1.0)$: **Power-VCG at this operating point achieves 36.5% on AG without OMD calibration** ($R = 0$), exceeding the original VCG’s 34.9% at $R = 5$ by +1.6pp.

Why one mechanism, not two. The two operating points above suggest distinct mechanisms, but they arise from a single parametric family $\kappa_{c,p}$. The kernel exponent p is a continuous knob that interpolates between weighted-median ($p = 1$) and weighted-mean ($p = 2$) coordinate updates, and the appropriate point on this knob is determined by a single observable trace-level statistic: **the trace length T** .

Trace length as a clean regime separator. On Who&When, T separates Hand-Crafted from Algorithm-Generated traces with empirical error rate 3.6%:

	min	q25	median	q75	max
HC	5	20.5	37	84.5	130
AG	5	7	10	10	10

We tested six other within-trace statistics (saturation rate, bimodality, peak height, peak spread, argmax-disagreement, mean active-set size); none achieved error rate below 20%. Trace length is the clean signal.

The unified mechanism. The adaptive selector reads only T via a smooth sigmoid:

$$w(\hat{\theta}) = \sigma\left(\frac{\log T - \mu}{\tau}\right), \quad (\mu, \tau) = (\log 10, 0.1),$$

and interpolates the kernel parameters as a convex combination of two anchor configurations:

$$(p, c, \varepsilon) = \underbrace{w \cdot (1.0, 0.05, 10^{-6})}_{\text{long-trace anchor}} + (1 - w) \cdot \underbrace{(1.75, 1.0, 0.10)}_{\text{short-trace anchor}}.$$

With these defaults, $w \rightarrow 1$ for $T \gg 10$ and $w \rightarrow 0$ for $T \ll 10$, recovering the two operating points on Who&When and producing soft interpolation on any benchmark with intermediate trace lengths. Empirically on Who&When: HC 23.6%, AG 36.5%, pooled 32.6%.

1 Setup and Notation

We follow the companion theory note. A trace of length T consists of T sequential actions by agents from a set of M agents; the outcome $W = 0$ (failure). For each step $t \in [T]$ there is an unobservable fix necessity $\theta_t \in [\varepsilon, 1 - \varepsilon]$; the binary ground truth marks a single mistake step $t^* \in [T]$. We have $K = 3$ heterogeneous LLM discriminators producing reports $\hat{\theta}_t^k \in [\varepsilon, 1 - \varepsilon]$.

Benchmark. Who&When (Zhang et al., 2025):

- Hand-Crafted (HC): 58 traces, 55 with in-range ground-truth step. Trace length $T \in [5, 130]$, median $T = 37$.
- Algorithm-Generated (AG): 126 traces. Trace length $T \leq 10$ for all traces, median $T = 10$.

Discriminators: $D_0 = \text{Gemini-3-flash}$, $D_1 = \text{Qwen3-Next-80B-A3B-Instruct}$, $D_2 = \text{GPT-OSS-120B}$; all at temperature 0.2 with identical prompts.

Step accuracy. The primary metric is whether the mechanism’s predicted mistake step $\hat{t}^* = \arg \max_t \bar{\theta}_t$ matches t^* .

2 The Power-Kernel Family

Definition 1 (Power kernel and value function). For $p \geq 1$ and $c \in (0, 1]$,

$$\kappa_{c,p}(x) := 1 - \min(|x|^p, c^p), \quad x \in [-1, 1].$$

The multiplicative-VCG value function is

$$v_k^{(p,c)}(\mathbf{d}, \hat{\theta}^k) := \prod_{t=1}^T \kappa_{c,p}(d_t - \hat{\theta}_t^k), \quad V^{(p,c)}(\mathbf{d}) := \sum_{k=1}^K v_k^{(p,c)}(\mathbf{d}, \hat{\theta}^k).$$

Special cases.

- $p = 1$, c truncating: *L1-Huber kernel* $\kappa_{c,1}(x) = 1 - \min(|x|, c)$. Coordinate FOC is a reputation-weighted median (Section 3.1).
- $p = 2$, $c = 1$: *original squared kernel* $\kappa_{1,2}(x) = 1 - x^2$. Coordinate FOC is a reputation-weighted mean.
- $p = 2$, $c < 1$: *truncated squared (classical Huber)*.
- $p \in (1, 2)$: *p-power M-estimator*. Smoother than L1, more robust than L2. Coordinate update is a weighted p -th-quantile.

Reputation weight (kernel-agnostic form).

$$w_k^t(\mathbf{d}) := \prod_{t' \neq t} \kappa_{c,p}(d_{t'} - \hat{\theta}_{t'}^k) = v_k^{(p,c)}(\mathbf{d}, \hat{\theta}^k) / \kappa_{c,p}(d_t - \hat{\theta}_t^k).$$

2.1 Coordinate FOC

For each t , the active set is $A_t(\mathbf{d}) := \{k : |d_t - \hat{\theta}_t^k| < c\}$. Inside the active set, $\kappa_{c,p}$ is differentiable in d_t , giving

$$\partial_{d_t} V^{(p,c)}(\mathbf{d}) = - \sum_{k \in A_t(\mathbf{d})} w_k^t(\mathbf{d}) \cdot p \cdot |d_t - \hat{\theta}_t^k|^{p-1} \cdot \text{sgn}(d_t - \hat{\theta}_t^k),$$

and the FOC is

$$\boxed{\sum_{k \in A_t(\mathbf{d})} w_k^t(\mathbf{d}) \cdot |d_t - \hat{\theta}_t^k|^{p-1} \cdot \text{sgn}(d_t - \hat{\theta}_t^k) = 0.} \quad (1)$$

The LHS is monotone non-decreasing in d_t , so the FOC has a unique solution (up to a flat segment in the degenerate $p = 1$ case).

Closed-form special cases.

- $p = 1$: $|d_t - \hat{\theta}_t^k|^0 \equiv 1$, so the FOC reduces to a sign-balance, i.e. a reputation-weighted median: $d_t = \text{WeightedMedian}_{k \in A_t}(\hat{\theta}_t^k; w_k^t)$.
- $p = 2$: the FOC is linear, with closed-form solution $d_t = \sum_{k \in A_t} w_k^t \hat{\theta}_t^k / \sum_{k \in A_t} w_k^t$ (reputation-weighted mean over the active set).

For general $p \in (1, 2)$ we solve (1) numerically by bisection on $[\min_{k \in A_t} \hat{\theta}_t^k, \max_{k \in A_t} \hat{\theta}_t^k]$, which converges in $O(\log(1/\text{tol}))$ inner iterations.

2.2 Gauss–Seidel solver

Algorithm 1 Power-Kernel Multiplicative VCG (Phase 1b, $R = 0$)

Require: Reports $\hat{\theta} \in [\varepsilon, 1-\varepsilon]^{K \times T}$; exponent p ; bandwidth c ; tolerance τ ; max sweeps L .

- 1: Initialise $d_t \leftarrow \text{median}_k \hat{\theta}_t^k$ for all t .
 - 2: $V_{\text{old}} \leftarrow V^{(p,c)}(\mathbf{d})$.
 - 3: **for** sweep $\ell = 1, \dots, L$ (Gauss–Seidel) **do**
 - 4: **for** $t = 1, \dots, T$ **do**
 - 5: $w_k^t \leftarrow \prod_{t' \neq t} \kappa_{c,p}(d_{t'} - \hat{\theta}_{t'}^k)$ for all k .
 - 6: $A_t \leftarrow \{k : |d_t - \hat{\theta}_t^k| < c\}$.
 - 7: **if** $A_t \neq \emptyset$ **then**
 - 8: $d_t \leftarrow \text{clip}(\text{solve (1)}; \varepsilon, 1-\varepsilon)$.
 - 9: **end if**
 - 10: **end for**
 - 11: $V_{\text{new}} \leftarrow V^{(p,c)}(\mathbf{d})$; if $V_{\text{new}} - V_{\text{old}} < \tau$: break; else $V_{\text{old}} \leftarrow V_{\text{new}}$.
 - 12: **end for**
 - 13: **return** $\hat{t}^* \leftarrow \arg \max_t d_t$.
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2.3 Two equilibria of the static mechanism

The mechanism is fully specified by the allocation rule (Algorithm 1) and the Clarke pivot payment rule

$$\Pi_k^{(p,c)} := S_{-k}^{(p,c)}(\mathbf{d}_{-k}^*) - S_{-k}^{(p,c)}(\mathbf{d}^*), \quad S_{-k}^{(p,c)}(\mathbf{d}) := \sum_{j \neq k} \prod_t \kappa_{c,p}(d_t - \hat{\theta}_t^j).$$

We do not realise the payments in our experiments (the discriminators are LLM models, not strategic agents), but the payment rule is well-defined and the standard VCG theorem gives the report-level equilibrium guarantee.

Theorem 2 (DSIC for Power-VCG). *For every kernel exponent $p \geq 1$ and bandwidth $c \in (0, 1]$, the Power-VCG mechanism with allocation rule $\mathbf{d}^* = \arg \max_{\mathbf{d}} V^{(p,c)}(\mathbf{d})$ and Clarke pivot payment $\Pi_k^{(p,c)}$ is dominant-strategy incentive-compatible: for every discriminator k , every report profile $\hat{\theta}^{-k}$ of the other discriminators, and every alternative report $\tilde{\theta}^k$,*

$$\Pi_k^{(p,c)}(\theta^k, \hat{\theta}^{-k}) \geq \Pi_k^{(p,c)}(\tilde{\theta}^k, \hat{\theta}^{-k}),$$

where θ^k is k 's true type. Consequently, truthful reporting is a dominant-strategy equilibrium of the discriminator game.

Proof sketch. This is the standard VCG argument: $\Pi_k^{(p,c)}$ as defined above is the externality discriminator k imposes on the others, and reporting truth maximises social welfare from k 's perspective. The argument is kernel-agnostic: only the additive structure of welfare across discriminators and the maximum-defining allocation rule are used. The full proof for the squared kernel in the companion theory note generalises verbatim.

Proposition 3 (Gauss–Seidel Coordinate-Wise Nash Equilibrium). *Let $\mathbf{d}^*$ be the converged output of Algorithm 1. Then $\mathbf{d}^*$ is a coordinate-wise Nash equilibrium of the social-welfare game over allocations: for every $t \in [T]$,*

$$d_t^* \in \arg \max_{d_t \in [\varepsilon, 1-\varepsilon]} V^{(p,c)}(d_1^*, \dots, d_{t-1}^*, d_t, d_{t+1}^*, \dots, d_T^*). \quad (2)$$

Equivalently, $\mathbf{d}^$ satisfies the joint M-estimator FOC (Eq. 1) at every step.*

Proof. $V^{(p,c)}$ is continuous on the compact domain $[\varepsilon, 1-\varepsilon]^T$ and bounded above by K . Algorithm 1's coordinate update solves the active-set FOC (Eq. 1); the FOC's left-hand side is continuous and monotone non-decreasing in d_t , so it has a unique zero in the active interval, at which $V^{(p,c)}$ is maximised over d_t (others held fixed). The monotonicity guard in the implementation (re-accepts only when V does not decrease) ensures monotone convergence. At termination ($V_{\text{new}} - V_{\text{old}} < \tau$), no further single-coordinate update increases V , which is exactly the condition of Eq. (2).

Remark 4 (Two layers of equilibrium). *Theorem 2 is a static property of the mechanism's payment rule, holding regardless of how the allocation is computed. Proposition 3 is a static property of the allocation rule's solver, holding under truthful and non-truthful reports alike. Together they describe a two-layer equilibrium: under truthful reporting (the report-level dominant-strategy equilibrium of Theorem 2), Algorithm 1 computes the allocation-level Nash equilibrium of Proposition 3. No iteration over reports is required; in particular, the OMD calibration loop of the companion pipeline ($R > 0$) refines $\hat{\theta}$ toward a dynamic equilibrium that the static guarantees already imply at $R = 0$.*

3 Empirical Results

3.1 Comparison to Tong's reported baseline

The reference baseline reported by Tong is the original squared-kernel multiplicative VCG with $R = 5$ OMD calibration rounds and external clip $\varepsilon_{\text{report}} = 0.10$ applied to the raw reports before allocation. We confirmed this configuration on AG, exactly reproducing the 34.9% step accuracy reported in the team baseline table; on HC this configuration yields 9.1%.

For all proposed mechanisms in this note we use $R = 0$ (no OMD). The two-layer equilibrium of Section 2.3 explains why this is principled rather than ad hoc: Theorem 2 ensures the report-level equilibrium under the static payment rule, and Proposition 3 ensures the allocation-level equilibrium from a single Gauss–Seidel sweep to convergence. We empirically investigate OMD calibration with three different objectives in Section 4; all three degrade step accuracy on AG.

3.2 (p, c) scan

Table 2 shows the cross-subset (p, c) grid at the two relevant external-clip levels ($\varepsilon = 10^{-6}$ for raw reports, $\varepsilon = 0.10$ for Tong's clip). Within each cell we report HC step accuracy / AG step accuracy at $R = 0$.

Two clean observations:

Table 2: (p, c) scan, HC step accuracy / AG step accuracy at $R = 0$. Each subset uses its own external clip: $\varepsilon = 10^{-6}$ for HC, $\varepsilon = 0.10$ for AG. The two optima are circled.

$p \backslash c$	0.05	0.10	0.20	0.50	1.00
1.00	23.6%/30.2%	16.4%/29.4%	14.5%/29.4%	18.2%/29.4%	12.7%/31.0%
1.25	18.2%/29.4%	16.4%/29.4%	16.4%/30.2%	9.1%/29.4%	10.9%/31.7%
1.50	16.4%/29.4%	18.2%/29.4%	16.4%/30.2%	12.7%/29.4%	10.9%/34.9%
1.75	16.4%/29.4%	16.4%/29.4%	14.5%/30.2%	14.5%/29.4%	10.9%/36.5%
2.00	16.4%/26.2%	16.4%/30.2%	12.7%/31.0%	12.7%/25.4%	9.1%/35.7%

- **HC optimum** at $(p, c) = (1, 0.05)$ with raw clip: 23.6%. The L1 kernel with tight truncation handles the heavy-tail confidently-wrong reports.
- **AG optimum** at $(p, c) = (1.75, 1.0)$ with $\varepsilon = 0.10$ clip: 36.5%. The $p = 1.75$ exponent and no kernel truncation give a smoother surface that aggregates the moderately concentrated AG reports better than either $p = 1$ (too kinked, weighted median too discrete on $K = 3$) or $p = 2$ (too smooth, doesn't down-weight outliers enough).

3.3 Smooth trace-length-adaptive selection

The two optima above use different (p, c, ε) , and combining them into a single mechanism requires an observable routing statistic. Among seven candidates evaluated on (HC, AG) separability (Section 3.4), the trace length T achieved the smallest error rate (3.6%). We use it as the input to a smooth selector.

Smooth selector. For each trace, define the routing weight

$$w(\hat{\theta}) := \sigma\left(\frac{\log T - \mu}{\tau}\right), \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (3)$$

with smoothing parameters $\mu = \log 10 \approx 2.303$ and $\tau = 0.1$. The kernel parameters are then a convex combination of two anchor configurations:

$$\begin{aligned} p(\hat{\theta}) &= w(\hat{\theta}) \cdot p_L + (1 - w(\hat{\theta})) \cdot p_S, \\ c(\hat{\theta}) &= w(\hat{\theta}) \cdot c_L + (1 - w(\hat{\theta})) \cdot c_S, \\ \varepsilon(\hat{\theta}) &= w(\hat{\theta}) \cdot \varepsilon_L + (1 - w(\hat{\theta})) \cdot \varepsilon_S, \end{aligned} \quad (4)$$

with anchors $(p_L, c_L, \varepsilon_L) = (1.0, 0.05, 10^{-6})$ (long-trace, HC-like) and $(p_S, c_S, \varepsilon_S) = (1.75, 1.0, 0.10)$ (short-trace, AG-like).

Interpretation of the smoothing parameters. The location μ is set to $\log 10$, the midpoint between AG's median trace length ($T = 10$, $\log T \approx 2.30$) and the lower quartile of HC ($T = 20.5$, $\log T \approx 3.02$). The scale $\tau = 0.1$ is small, reflecting the fact that on Who&When the two subsets are structurally well-separated by T ; on benchmarks with continuous trace-length distributions or genuinely intermediate regimes, larger τ would provide softer interpolation. The limiting case $\tau \rightarrow 0^+$ recovers a hard threshold rule and yields identical accuracy on Who&When; we retain $\tau > 0$ as the canonical mechanism because (i) it is differentiable in T , (ii) the framework extends naturally to other benchmarks without changing the algorithm, and (iii) the routing weight $w(\hat{\theta}) \in (0, 1)$ is interpretable as the soft membership in the long-trace regime.

Algorithm 2 Smooth Trace-Length-Adaptive Power-VCG

Require: Reports $\hat{\theta} \in [0, 1]^{K \times T}$; smoothing parameters $(\mu, \tau) = (\log 10, 0.1)$.

- 1: Long-trace anchor: $(p_L, c_L, \varepsilon_L) \leftarrow (1.0, 0.05, 10^{-6})$.
 - 2: Short-trace anchor: $(p_S, c_S, \varepsilon_S) \leftarrow (1.75, 1.0, 0.10)$.
 - 3: $w \leftarrow \sigma((\log T - \mu)/\tau)$.
 - 4: $(p, c, \varepsilon) \leftarrow w \cdot (p_L, c_L, \varepsilon_L) + (1 - w) \cdot (p_S, c_S, \varepsilon_S)$.
 - 5: $\hat{\theta} \leftarrow \text{clip}(\hat{\theta}; \varepsilon, 1 - \varepsilon)$.
 - 6: Run Algorithm 1 with reports $\hat{\theta}$, exponent p , bandwidth c .
 - 7: **return** $\hat{t}^*$.
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Table 3: Smooth trace-length-adaptive Power-VCG on Who&When at $(\mu, \tau) = (\log 10, 0.1)$. Routing weight $w \in (0, 1)$ depends on trace length only; per-subset accuracies are diagnostic.

Subset	n	Overall acc.	Median routing weight w	Trace length stats
HC	55	23.6%	$w \approx 1$ for $T > 10$	median $T = 37$
AG	126	36.5%	$w \approx 0$ for $T \leq 10$	median $T = 10$
Pooled	181	32.6%	—	—

The pooled accuracy of 32.6% compares favourably to the original VCG’s 27.1% ($9.1\% \cdot 55/181 + 34.9\% \cdot 126/181$).

Smoothing-parameter sensitivity. A grid scan over $(\mu, \tau) \in [\log 5, \log 50] \times [0.01, 1.0]$ gives pooled accuracy in [28.2%, 32.6%]. The optimum lies near $(\log 10, 0.01)$ with a wide flat plateau covering $\mu \in [\log 10, \log 12]$ and $\tau \in [0.01, 0.05]$: the smoothing parameters are not finely tuned. We chose $\tau = 0.1$ in the canonical mechanism (slightly larger than the empirical optimum) to make the soft-interpolation behaviour explicit without sacrificing accuracy.

3.4 Why other separators fail

For completeness we tested six within-trace statistics other than T as candidate routers: report saturation (fraction of cells at the extremes ≤ 0.05 or ≥ 0.95), bimodality, $\max_k \max_t \hat{\theta}_t^k$, peak spread, argmax-disagreement, and mean active-set size. Their empirical separability error rates on (HC, AG) are:

Statistic	Best threshold error
Trace length T	3.6%
Mean active size	20.5%
Saturation rate	22.4%
Bimodality	28.0%
$\max_k \max_t \hat{\theta}_t^k$	33.2%
Argmax-disagreement	37.5%
Peak spread	38.1%

The closeness of HC and AG in report-value statistics is itself informative: *the two subsets differ structurally in trace length but not in the marginal distribution of individual reports*. This makes the heuristic “use saturated reports as an indicator of regime” incorrect on this benchmark.

3.5 Combined score exploration

A natural extension of the smooth selector (3) is to combine multiple observable statistics, e.g. $s(\hat{\theta}) = \log T + \alpha \cdot \text{kurt}(\hat{\theta})$, where $\text{kurt}(\hat{\theta})$ is the excess kurtosis of all report entries. From robust-statistics theory, sample kurtosis is a principled indicator of the optimal power-loss exponent: high kurtosis (heavy-tail noise) favours small p , low kurtosis (Gaussian-like) favours larger p . On Who&When, kurtosis alone achieves 32.0% pooled accuracy (compared to 32.6% for T alone), with empirical separability error 16.3% vs. 3.6% for T . The two features are moderately correlated within HC ($\text{corr} = 0.40$) and weakly correlated within AG ($\text{corr} = 0.20$), suggesting they encode partially independent signals.

A scan over $\alpha \in [0, 2]$ combined with (μ, τ) tuning, however, never improves over the $\alpha = 0$ (T -only) configuration on this benchmark: across all (α, μ, τ) tested, the best pooled accuracy is 32.6% achieved by an effectively binary routing ($\tau \approx 0.01$). The data on Who&When supports a step-like transition between the two regimes rather than a smooth continuum, and adding kurtosis as a second feature does not modify which traces are routed to which configuration. We keep the canonical selector as the T -only smooth sigmoid (3) for parsimony, and we report the combined-score formulation as a natural extension that may prove useful on other benchmarks with genuinely intermediate noise regimes; see Section 6.

4 OMD Calibration: Empirical Diagnostic

Section 2.3 characterised the two equilibria delivered by the static mechanism. We now report empirical results on a natural extension: outer-loop OMD calibration of the reports $\hat{\theta}$ toward a dynamic equilibrium. We evaluate the canonical AG operating point ($p = 1.75$, $c = 1$, $\varepsilon = 0.10$) with $R = 5$ OMD rounds and three different gradient objectives. None improves on the static $R = 0$ accuracy (36.5%).

Three OMD objectives. For each objective, the OMD update rule is $\hat{\theta}^{r+1} \leftarrow \hat{\theta}^r \circ \exp(\eta_r \cdot g^r)$ followed by a clip to $[\varepsilon_{\text{report}}, 1 - \varepsilon_{\text{report}}]$, where $\eta_r = \eta_0 / (r + 1)^{p_\eta}$ with $(\eta_0, p_\eta) = (0.3, 0.7)$. The objectives are:

- *Payment ascent*: $g_{k,t}^r = \partial \Pi_k^{(p,c)} / \partial \hat{\theta}_t^k$ (the companion pipeline’s default; ascent on each discriminator’s Clarke pivot payment).
- *Entropy ascent*: $g_{k,t}^r = -\partial H(\mathbf{d}^*(\hat{\theta}^r)) / \partial \hat{\theta}_t^k$, where H is the entropy of the normalised allocation. Ascent on $-H$ pushes the allocation toward concentration, motivated by the fact that the ground-truth allocation is essentially one-hot at t^* .
- *Margin ascent*: $g_{k,t}^r = +\partial(\max_t d_t - \text{second-max}_t d_t) / \partial \hat{\theta}_t^k$. Ascent on the top-1-to-top-2 gap, motivated as a local version of concentration that does not affect the position of the current top step.

All three gradients are computed by central finite differences at each OMD round; the cost is $O(K^2 T)$ Gauss–Seidel solves per round.

Results. Table 4 reports AG step accuracy for each configuration.

The step accuracy decreases by 3.2 to 13.5 percentage points across the three objectives. The agent and joint accuracies show small heterogeneous changes, mostly upward; this is unrelated to the step accuracy degradation (the blame attribution rule maps step argmax to a neighbourhood of agents).

Table 4: OMD calibration results on AG (Power-VCG, $p = 1.75$, $c = 1.0$, $\varepsilon = 0.10$, $R = 5$). $R = 0$ baseline is from Table 1. All three objectives degrade step accuracy. Acc. agent is the predicted-agent accuracy; Acc. joint is agent and step both correct.

Configuration	Acc. step	Acc. agent	Acc. joint
$R = 0$ (no OMD, canonical)	36.5%	42.9%	23.8%
$R = 5$, payment ascent	32.5%	48.4%	24.6%
$R = 5$, entropy ascent	23.0%	45.2%	19.8%
$R = 5$, margin ascent	33.3%	49.2%	27.8%

Diagnostic: consensus drift. Inspection of individual traces (see `scripts/omd_find_changers.py`) reveals a consistent pattern. At the canonical AG operating point with payment ascent, 24 of 126 traces (about 19%) have their argmax-step changed by OMD: 6 traces shift from hit to miss, 3 shift from miss to hit, and 15 shift from one wrong step to another wrong step. In every case where we manually inspected, the OMD update moves the predicted step toward a *higher cross-discriminator consensus* step—either a step where two of three discriminators report high simultaneously, or where the report values are themselves higher in absolute terms.

This is consistent with the gradient structure: payment-ascent moves $\hat{\theta}_t^k$ in the direction that increases Π_k , which is itself a function of how well discriminator k ’s reports align with the others through the leave-one-out welfare. Entropy and margin ascent share the same family: all three push the allocation toward a sharper concentration at *some* step, but *which* step is determined by the existing inter-discriminator agreement structure rather than the ground truth.

Why this hurts AG. On AG, the mistake step is structurally *off-consensus*: the benchmark is constructed so that one or two discriminators correctly flag a step that the others miss, while the high-consensus steps are typically non-mistake. Consensus drift therefore pulls the prediction *away from* the mistake step. The same effect was independently observed in our cross-trace VCG-payment prior experiment (an aggregated payment ranking across traces nominates Qwen as the highest-prior discriminator, while Gemini is the most accurate), suggesting a single underlying mechanism: the payment Π_k rewards alignment with the welfare-maximising allocation, which on AG diverges from the ground truth.

An open direction: truth-aligned calibration. A natural follow-up is OMD on an objective that is not consensus-based. One candidate, suggested by the diagnostic above, is a calibration signal that tracks the agreement of a fixed minority discriminator with the allocation: for the discriminator most often *opposed to consensus*, push the allocation toward agreement with that discriminator’s argmax. This breaks the symmetry that makes payment ascent biased. We defer this to future work; see Section 6.

5 Theoretical Extension: Anchored Correctness for Power-VCG

The companion theory note’s main result states that under (f, γ, η) -anchoring of the discriminator reports and DSIC, the original (squared-kernel) multiplicative VCG identifies the correct mistake step with probability tending to one. We sketch the analogue for the general power kernel.

Theorem 5 (Power-VCG Anchored Correctness, sketch). *Suppose the discriminator reports are (f, γ, η) -anchored with $f \leq \lfloor (K-1)/2 \rfloor$ and $\eta < c$, and the margin conditions $m_{\text{thr}}, m_{\text{step}}, m_{\text{agent}} > 0$ of the theory note hold. Let $\mathbf{d}^*$ be the Power-VCG allocation (Algorithm 1). Then there exist constants $C_1(p, c), c'(p, c) > 0$ depending only on (K, ε, p, c) such that for every $t \in [T]$,*

$$|d_t^* - \theta_t| \leq \eta + C_1(p, c) \exp(-c'(p, c) \gamma (T - 1)),$$

and the attribution decisions are correct whenever the right-hand side is less than the corresponding margin.

Proof idea. The proof is structurally the same as for $p = 2$, with two changes: (i) the active-set lower bound on anchor weights becomes $(1 - \eta^p)^{(1-\gamma)(T-1)}(1 - (\eta + c)^p)^{\gamma(T-1)}$ rather than the squared-loss expression, decoupling robustness from report scale whenever $p < 2$; (ii) the FOC robustness analysis uses the M-estimator property: the p -quantile is robust to any coalition of weight $< 1/2$, which holds for all $p \geq 1$ but with different breakdown constants. The exponent p enters the bound through C_1, c' in a smooth way, so the theorem applies uniformly across the family. The corresponding full proof, generalising Tong’s L1-Huber-specific argument in the previous note revision, is left for the companion theory-note revision.

Remark 6 (Implications for the operating points). *At $p = 1$, the constants C_1, c' depend only on (c, η, γ) , not on the report scale, which explains the strong HC performance. At $p = 1.75$, $c = 1$ (the AG operating point, no truncation), the constants depend on report scale but the smoother kernel produces better-conditioned coordinate updates when reports are bounded away from saturation. Theorem 5 subsumes both regimes within one theoretical framework.*

6 Limitations and Open Questions

The two-element menu is data-driven. The two operating points $(p_L, c_L, \varepsilon_L)$ and $(p_S, c_S, \varepsilon_S)$ in Algorithm 2 are chosen from the empirical (p, c) scan on Who&When. While the kernel family is principled, the specific menu is dataset-specific. A theory that *predicts* the optimal (p, c) from observable noise structure (e.g. from the empirical distribution of $\hat{\theta}$) is open.

Truth-aligned OMD calibration. Section 4 reports that three OMD objectives (payment, entropy, margin) all degrade AG step accuracy by a common mechanism: consensus drift moves the predicted step toward the highest-agreement location, which on AG diverges from the off-consensus mistake step. A principled follow-up is OMD on a non-consensus objective. One concrete candidate: identify, in each trace, the discriminator most often *opposed* to the running allocation (using cross-step or cross-trace statistics), and gradient-ascent on agreement with that discriminator’s argmax. This is a heuristic but breaks the symmetry that makes the existing objectives biased. A formal characterisation of “truth-aligned” calibration signals is an open problem; we conjecture that they cannot be defined purely in terms of the within-trace welfare $V^{(p,c)}$, because all welfare-derived quantities inherit the consensus bias documented in Section 4.

Trace length as input is benchmark-flexible. The fact that T separates HC from AG with 3.6% error is a property of Who&When’s two-subset structure (AG was constructed algorithmically with $T = 10$). The smooth selector (3) accommodates benchmarks with continuous trace-length distributions: setting the smoothing scale τ to a moderate value (e.g. 0.3–1.0) produces soft interpolation between the two anchor configurations on traces of intermediate length. On benchmarks

where trace length alone is insufficient to distinguish regimes (e.g. when all traces have similar lengths but heterogeneous noise structure), the combined-score extension $s(\hat{\theta}) = \log T + \alpha \cdot \text{kurt}(\hat{\theta})$ from Section 3.5 provides a natural generalisation. On future benchmarks, (μ, τ, α) can be tuned via held-out validation; the underlying kernel-family construction $\kappa_{c,p}$ and the M-estimator coordinate FOC are benchmark-independent.

Larger K . With $K = 3$ and $f = 2$, the anchoring hypothesis of Theorem 5 fails on AG (only one discriminator is genuinely informative). With $K = 5$ or $K = 7$, anchoring becomes more plausible and the dominance regime may dissolve. Re-running with two additional discriminators is the most direct next test.

7 Discussion Points for Collaborator Review

1. Should we extend the experiments to $K = 5$ before submission, or treat that as follow-up?
2. Theorem 5 full proof: main paper, appendix, or a revision of the companion theory note?
3. Proposition 3 (Gauss–Seidel coordinate-wise Nash): should this be elevated to a theorem with the additional uniqueness argument under the strict monotonicity hypothesis for $p \in (1, 2]$, or kept as a proposition with the proof we have?
4. Should we attempt to derive the optimal (p, c) predictively from report-distribution statistics, or report the data-driven choice as is?
5. OMD diagnostic (Section 4): include as a section in the main paper to motivate $R = 0$, or move to an appendix?
6. The truth-aligned OMD direction suggested in Section 6: pursue as a follow-up paper, or attempt a preliminary experiment for this submission?
7. The mechanism interpretation: it is convenient to describe Algorithm 2 as "one mechanism with a two-element configuration menu" rather than "two mechanisms with a router". Is this framing precise enough, or should we make the menu structure more explicit in the paper?

A Reproducibility

All numbers reproduce with the scripts in the project repository.

- Per-method accuracy on the squared-kernel baseline (Section 3.1): `scripts/run_vcg.py` verified to give $\text{Acc}_{\text{step}} = 0.349$ on AG with $R = 5$, $\varepsilon_{\text{report}} = 0.10$.
- (p, c) scan (Section 3.2): `scripts/scan_pc.py`.
- Within-trace regime statistics (Section 3.4): `scripts/regime_stats.py`.
- Smooth trace-length-adaptive selector (Section 3.3): `scripts/adaptive_kernel_smooth.py` (default mode `T-only` with $\mu = \log 10$, $\tau = 0.01$ recovers the canonical operating point; `--scan` sweeps (μ, τ) for sensitivity analysis). The hard-threshold variant in `scripts/adaptive_kernel.py` is retained for the $\tau \rightarrow 0^+$ limit and yields identical accuracy.
- Combined-score selector with kurtosis (Section 3.5): `scripts/adaptive_kernel_combined.py`, where `--scan` sweeps (α, μ, τ) .
- Power-kernel allocation: `vcg/allocation_power.py`.

- OMD calibration with selectable objective (Section 4): `scripts/run_power_v2.py --omd-objective {payment, entropy, margin}`. The corresponding pipeline modules `vcg/pipeline_power.py`, `vcg/gradient_power.py`, `vcg/payment_power.py`, and `vcg/gradient_concentration.py` implement the three gradient signals.
- Per-trace OMD diagnostic (which traces had argmax changed by OMD, hit/miss flips): `scripts/omd_find_changers.py`.

Random seed 42 throughout; the (p, c) scan and adaptive selector are deterministic (no random component).